

Assignment 2: Setup a Test Case

In this assignment, a 2D axisymmetric turbulent jet simulation was developed using SU2. The workflow included generating the computational mesh using Gmsh, preparing the configuration file, running the solver, and analyzing the results using ParaView.

The flow was modeled as a steady, compressible turbulent jet using the Reynolds-Averaged Navier-Stokes (RANS) equations with the SST turbulence model. The simulation was carried out for a Mach number of 0.2 and a Reynolds number of 2×10^6 under axisymmetric conditions.

Configuration Details:

Inlet:

- Total Temperature: 302.4 K
- Total Pressure: 118309.784 Pa
- Flow Direction: (1, 0, 0)
- Turbulence Intensity: 5%
- Eddy Viscosity Ratio: 15

Outlet:

- Static Pressure: 115056 Pa

Wall:

- Zero heat flux condition (adiabatic wall)

Symmetry:

- Axisymmetric boundary applied along the centerline.

Results:

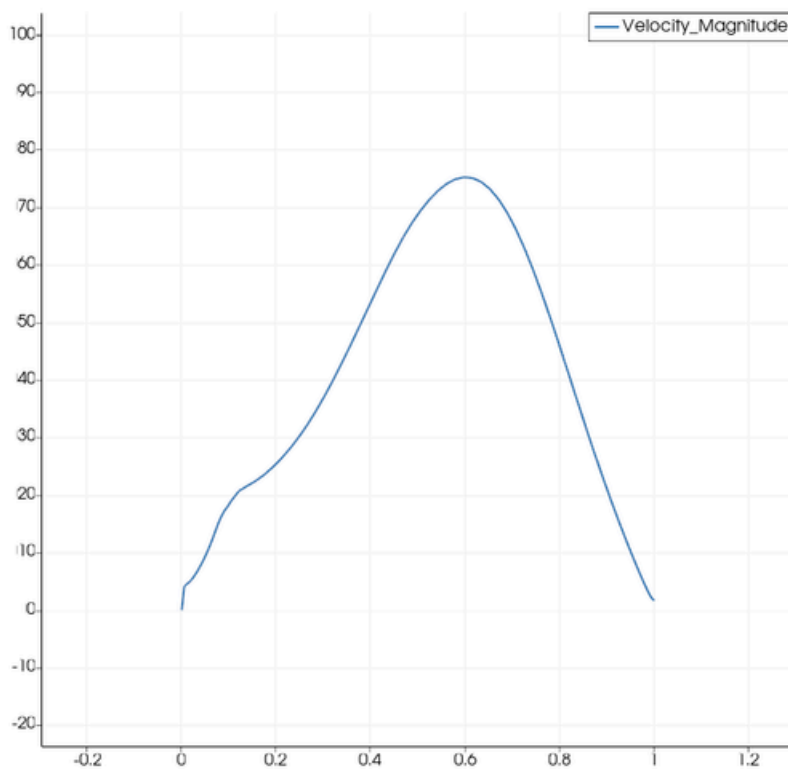
- The simulation results were visualized using ParaView.
- The velocity contours show a high-velocity region near the inlet, with changes in velocity distribution as the jet moves downstream due to mixing with the surrounding fluid.
- The temperature distribution indicates mixing between the jet and surrounding fluid, leading to gradual variation across the domain.

- Velocity profiles were extracted at different axial locations to examine the radial variation of velocity within the jet. The profiles show higher velocity near the central region and lower values towards the outer region, indicating the jet structure.
- These observations are consistent with the expected behavior of turbulent jet flows.

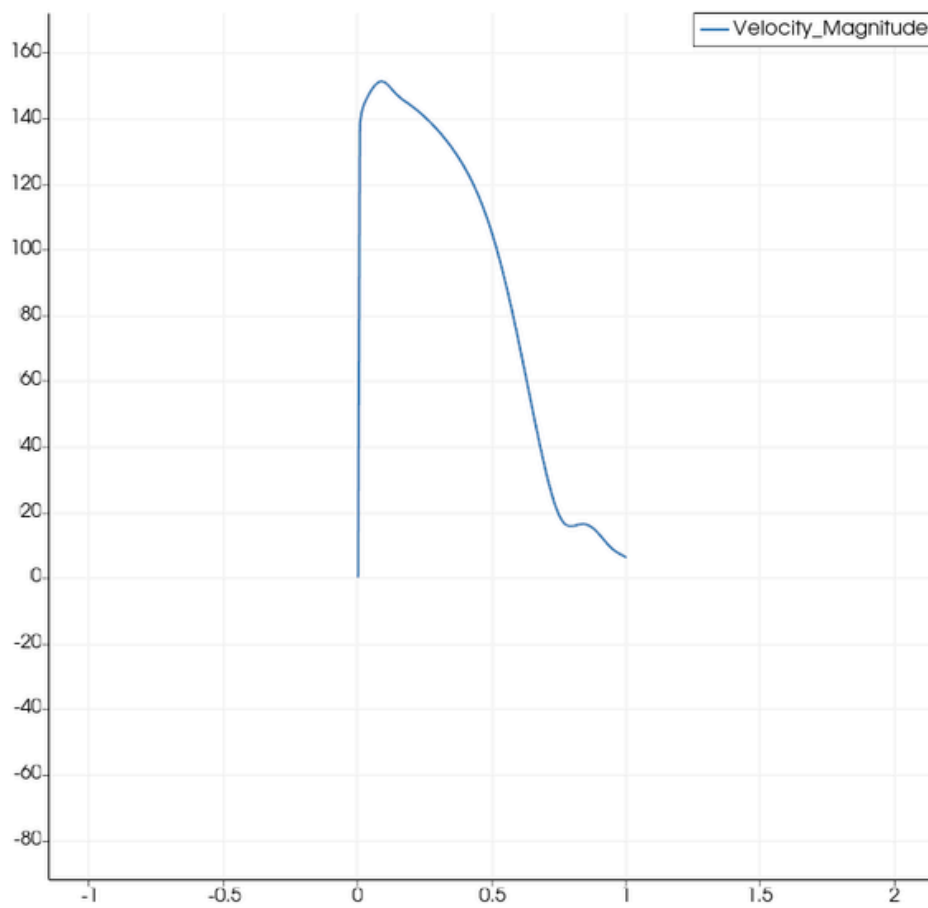
Conclusion:

The simulation successfully demonstrates the behavior of a turbulent jet, including velocity decay and radial spreading. This exercise helped in understanding the end-to-end process of CFD simulation using SU2, from mesh generation to result analysis.

Outputs:



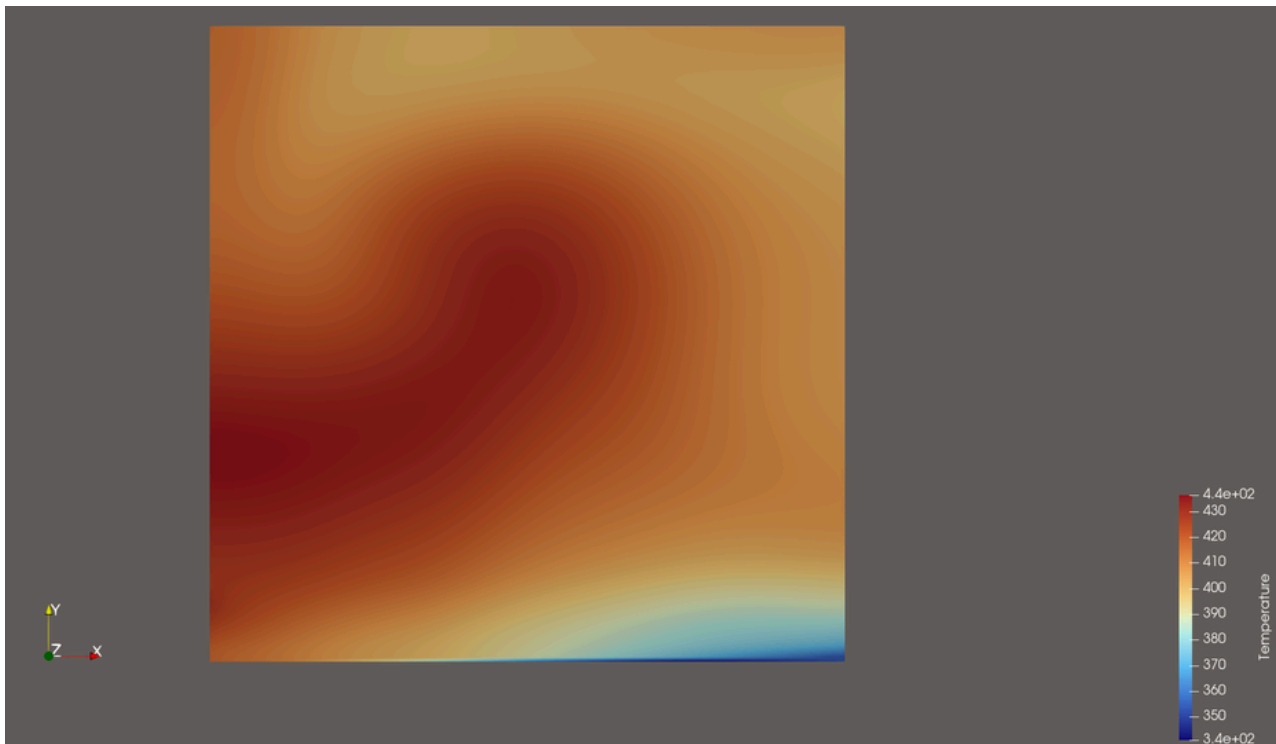
Velocity profile at $x = 0.05$



Velocity profile at $x = 0.5$

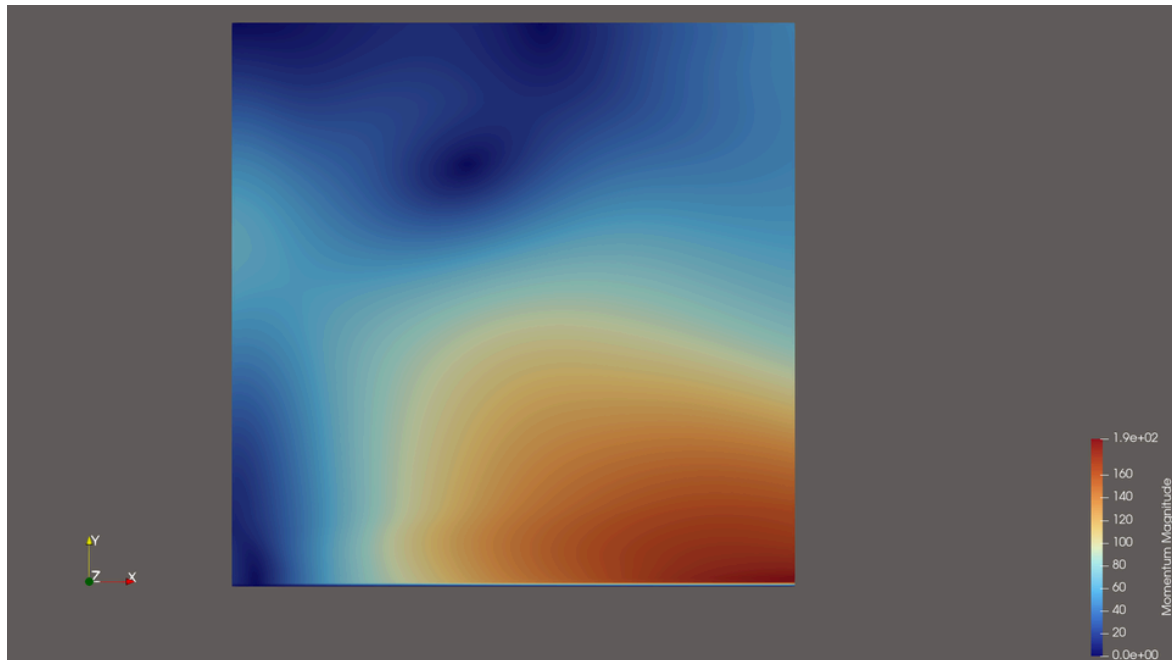
Temperature:

Temperature contour showing variation due to mixing in the flow.



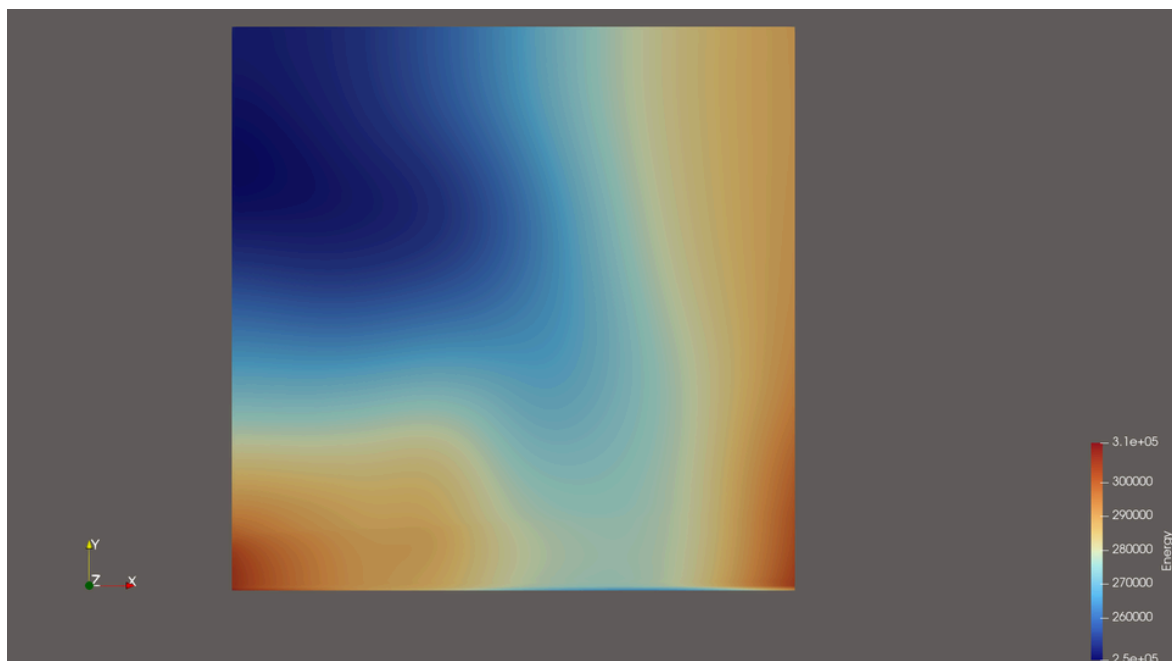
Momentum Magnitude:

Momentum magnitude contour representing the flow strength within the domain.



Energy:

Energy contour showing distribution of energy in the flow field.



Velocity:

Velocity magnitude contour showing the distribution of velocity in the jet flow.

